

TE Curve Verification Pipeline Familywise ONNX Report - wave3_3_raw_centered_shape_curve_aware

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml1`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory:
`output/validation_checks/track2_familywise_onnx_report/wave3_3_raw_centered_shape_curve_aware/2026-07-12-00-15-41__track2_wave3_3_raw_centered_shape_curve_aware_familywise_onnx_report ;`
- summary YAML:
`output/validation_checks/track2_familywise_onnx_report/wave3_3_raw_centered_shape_curve_aware/2026-07-12-00-15-41__track2_wave3_3_raw_centered_shape_curve_aware_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml ;`
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/wave3_3_raw_centered_shape_curve_aware/2026-07-12-00-15-41__track2_wave3_3_raw_centered_shape_curve_aware_familywise_onnx_report/model_inventory.csv ;`
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/wave3_3_raw_centered_shape_curve_aware/2026-07-12-00-15-41__track2_wave3_3_raw_centered_shape_curve_aware_familywise_onnx_report/per_curve_metrics.csv .`

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `wave3_3_raw_centered_shape_curve_aware` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_wave3_3_raw_centered_shape_curve_aware_fw_simplified_setpoints</code>	2026-07-11-16-12-47__te_wave3_3_raw_centered_shape_curve_aware_fw_simplified_setpoints	<code>simplified_curve_v1</code>
backward	<code>te_wave3_3_raw_centered_shape_curve_aware_bw_simplified_setpoints</code>	2026-07-11-16-42-36__te_wave3_3_raw_centered_shape_curve_aware_bw_simplified_setpoints	<code>simplified_curve_v1</code>
global	<code>te_wave3_3_raw_centered_shape_curve_aware_global_simplified_setpoints</code>	2026-07-11-15-43-36__te_wave3_3_raw_centered_shape_curve_aware_global_simplified_setpoints	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/wave3_3_raw_centered_shape_curve_aware/forward/2026-07-11-16-12-47__te_wave3_3_raw_centered_shape_curve_aware_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave3_3_raw_centered_shape_curve_aware/forward/2026-07-11-16-12-47__te_wave3_3_raw_centered_shape_curve_aware_fw_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=119-val_mae=0.00356710.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/wave3_3_raw_centered_shape_curve_aware/backward/2026-07-11-16-42-36__te_wave3_3_raw_centered_shape_curve_aware_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave3_3_raw_centered_shape_curve_aware/backward/2026-07-11-16-42-36__te_wave3_3_raw_centered_shape_curve_aware_bw_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=122-val_mae=0.00357800.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/wave3_3_raw_centered_shape_curve_aware/global/2026-07-11-15-43-36__te_wave3_3_raw_centered_shape_curve_aware_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave3_3_raw_centered_shape_curve_aware/global/2026-07-11-15-43-36__te_wave3_3_raw_centered_shape_curve_aware_global_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=114-val_mae=0.00357026.ckpt</code>

Aggregate Metrics

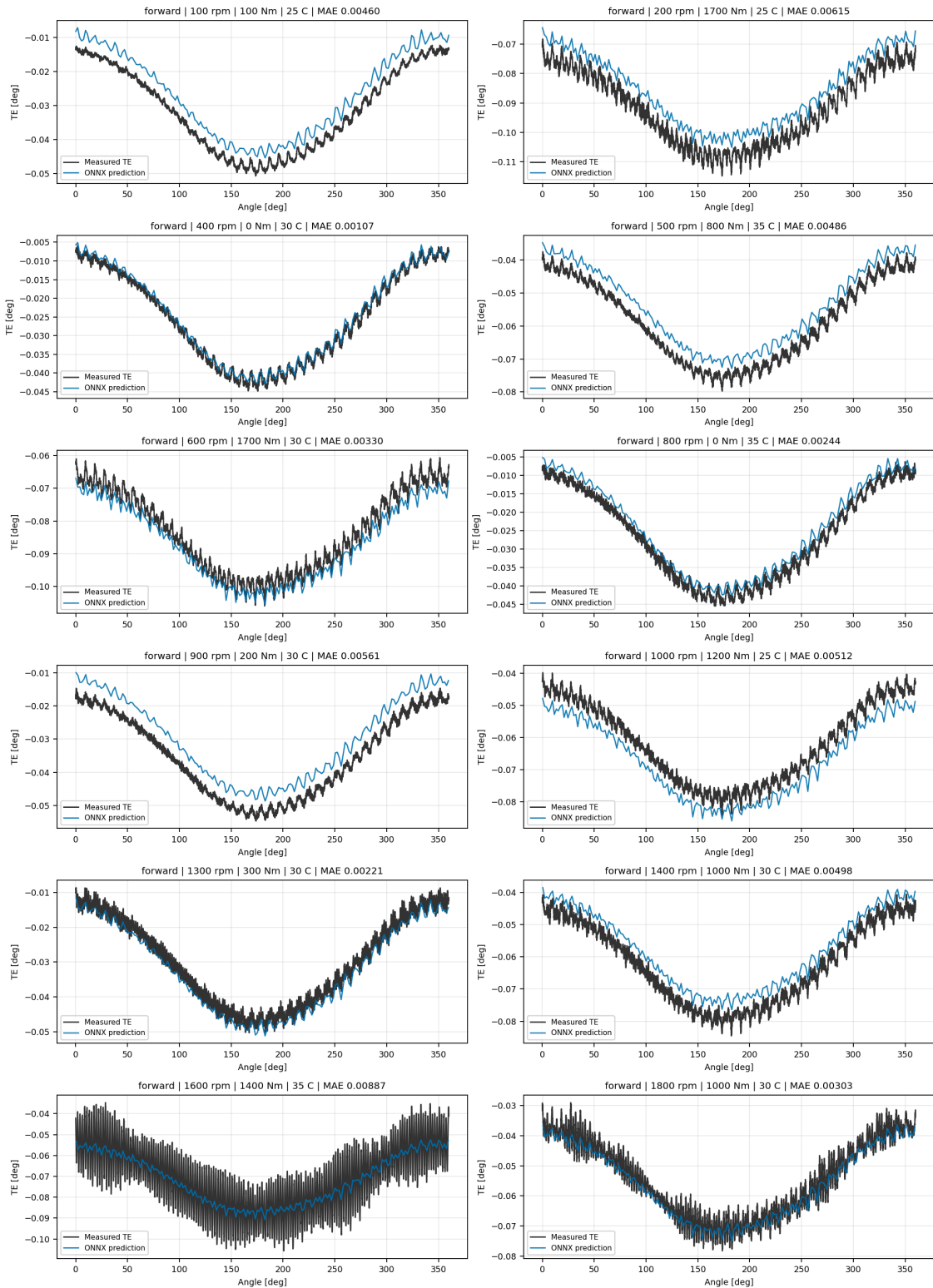
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003391	0.003667	7.934	13.954
backward	97	0.003556	0.003916	8.248	13.783
global	194	0.003524	0.003823	8.223	15.624

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000339	0.002961	0.001296	0.004132
backward	-0.000764	0.002942	0.001666	0.002972
global	0.000245	0.003026	0.001418	0.003109

Forward 12-Curve Page

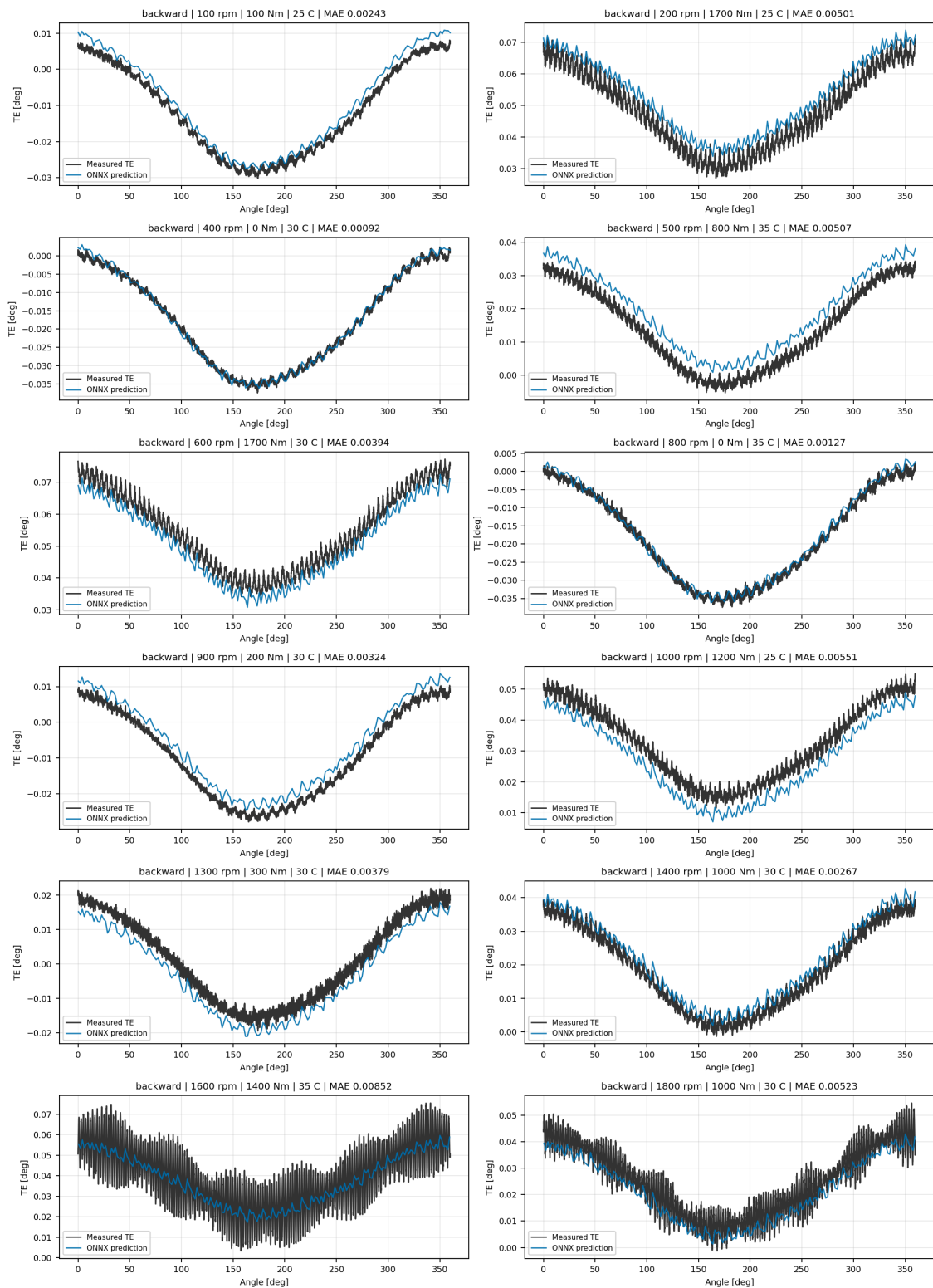
Simplified Dataset + Setpoints | wave3_3_raw_centered_shape_curve_aware | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

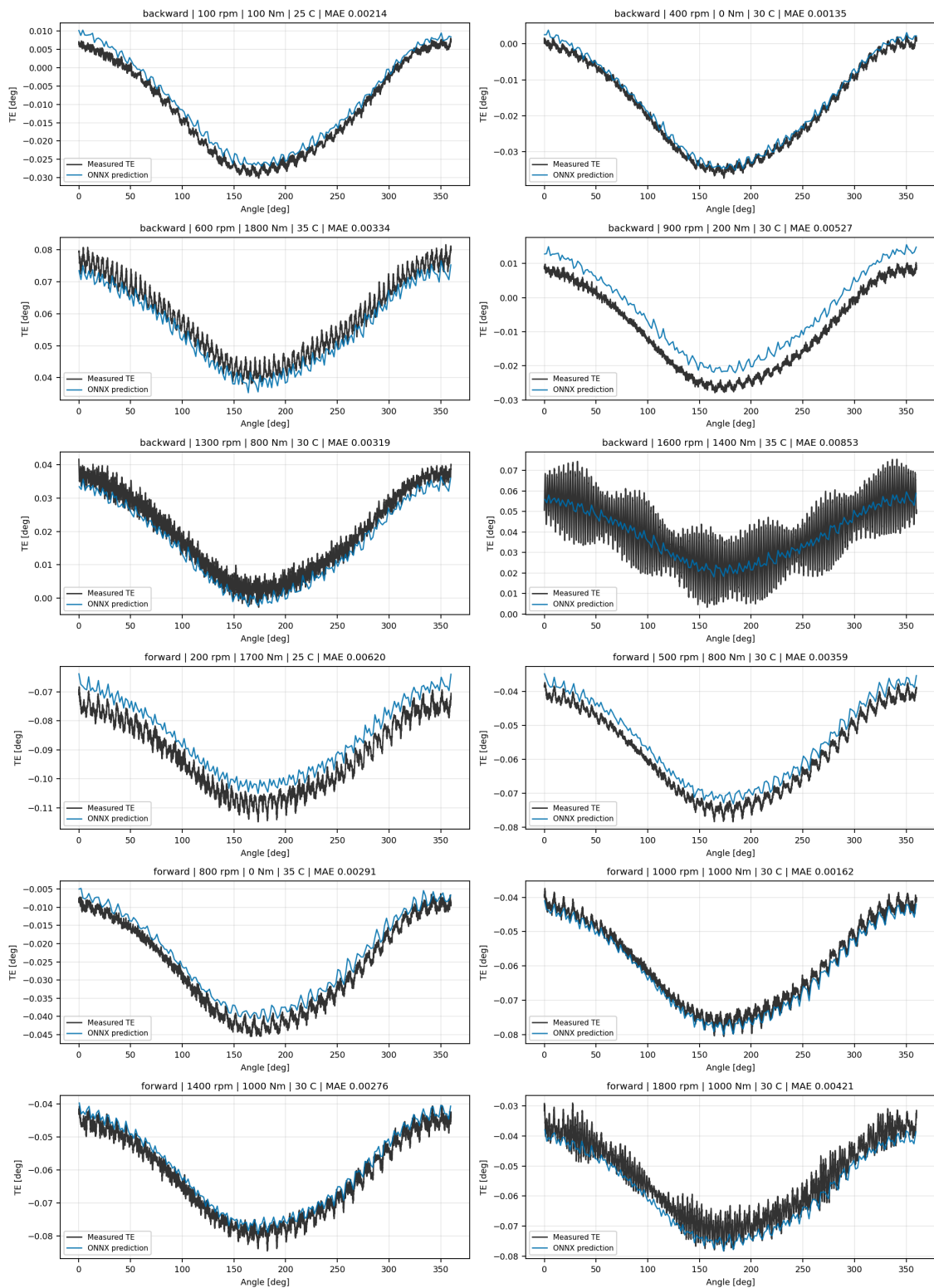
Simplified Dataset + Setpoints | wave3_3_raw_centered_shape_curve_aware | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | wave3_3_raw_centered_shape_curve_aware | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: wave3_3_raw_centered_shape_curve_aware ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave3_3_raw_centered_shape_curve_aware_fw_polished_setpoints	2026-07-11-18-01-02_te_wave3_3_raw_centered_shape_curve_aware_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_wave3_3_raw_centered_shape_curve_aware_bw_polished_setpoints	2026-07-11-18-25-23_te_wave3_3_raw_centered_shape_curve_aware_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_wave3_3_raw_centered_shape_curve_aware_global_polished_setpoints	2026-07-11-17-26-29_te_wave3_3_raw_centered_shape_curve_aware_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/wave3_3_raw_centered_shape_curve_aware/forward/2026-07-11-18-01-02_te_wave3_3_raw_centered_shape_curve_aware_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave3_3_raw_centered_shape_curve_aware/forward/2026-07-11-18-01-02_te_wave3_3_raw_centered_shape_curve_aware_fw_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=068-val_mae=0.00194145.ckpt
backward	models/polished_dataset/setpoints/exported/wave3_3_raw_centered_shape_curve_aware/backward/2026-07-11-18-25-23_te_wave3_3_raw_centered_shape_curve_aware_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave3_3_raw_centered_shape_curve_aware/backward/2026-07-11-18-25-23_te_wave3_3_raw_centered_shape_curve_aware_bw_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=121-val_mae=0.00193931.ckpt
global	models/polished_dataset/setpoints/exported/wave3_3_raw_centered_shape_curve_aware/global/2026-07-11-17-26-29_te_wave3_3_raw_centered_shape_curve_aware_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave3_3_raw_centered_shape_curve_aware/global/2026-07-11-17-26-29_te_wave3_3_raw_centered_shape_curve_aware_global_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=112-val_mae=0.00195150.ckpt

Aggregate Metrics

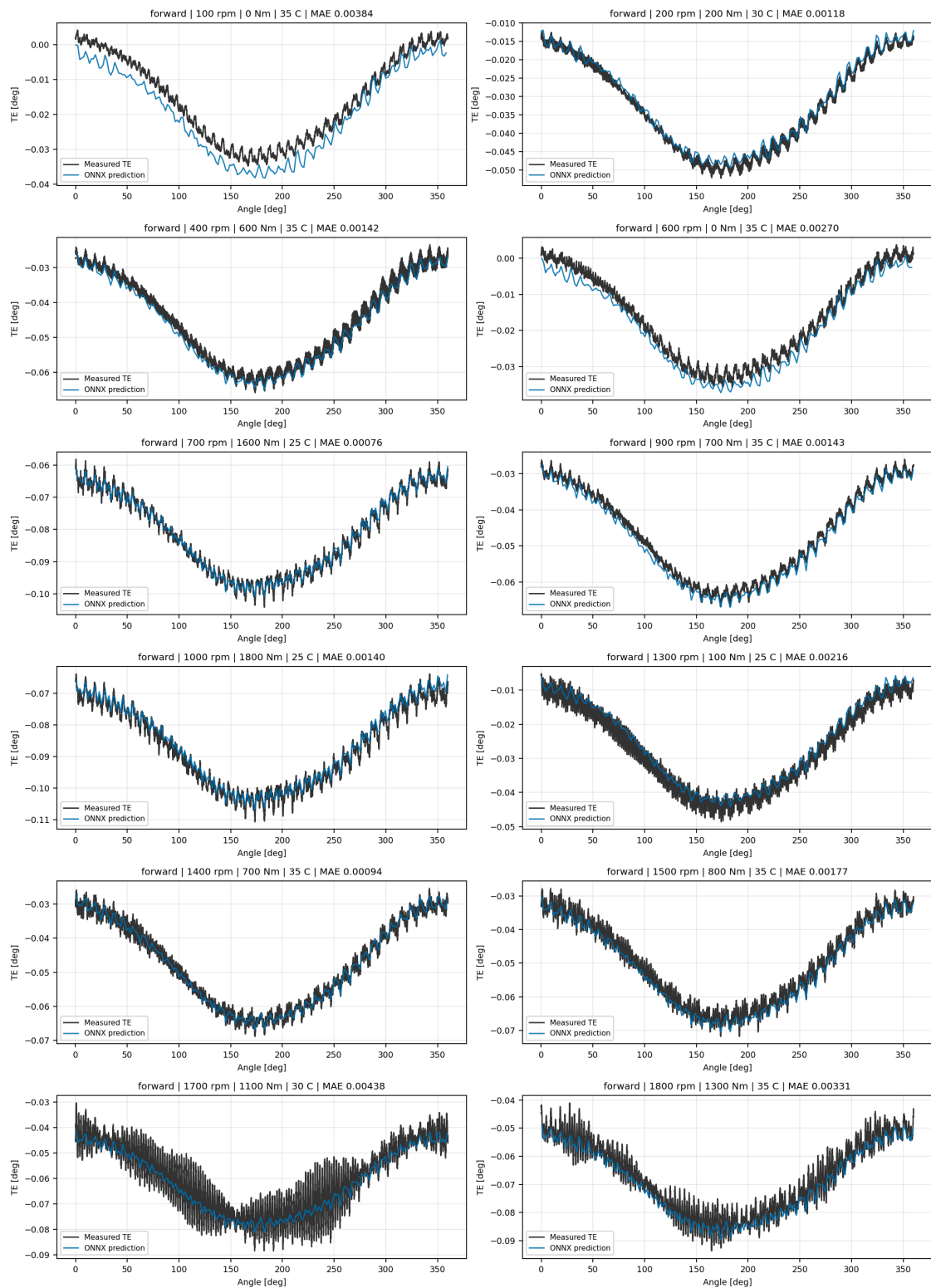
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002039	0.002414	4.507	11.166
backward	94	0.002539	0.002979	4.657	10.709
global	194	0.002271	0.002680	4.563	10.618

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000251	0.001152	0.001577	0.003558
backward	-0.000085	0.001050	0.002095	0.005563
global	-0.000113	0.001000	0.001866	0.004913

Forward 12-Curve Page

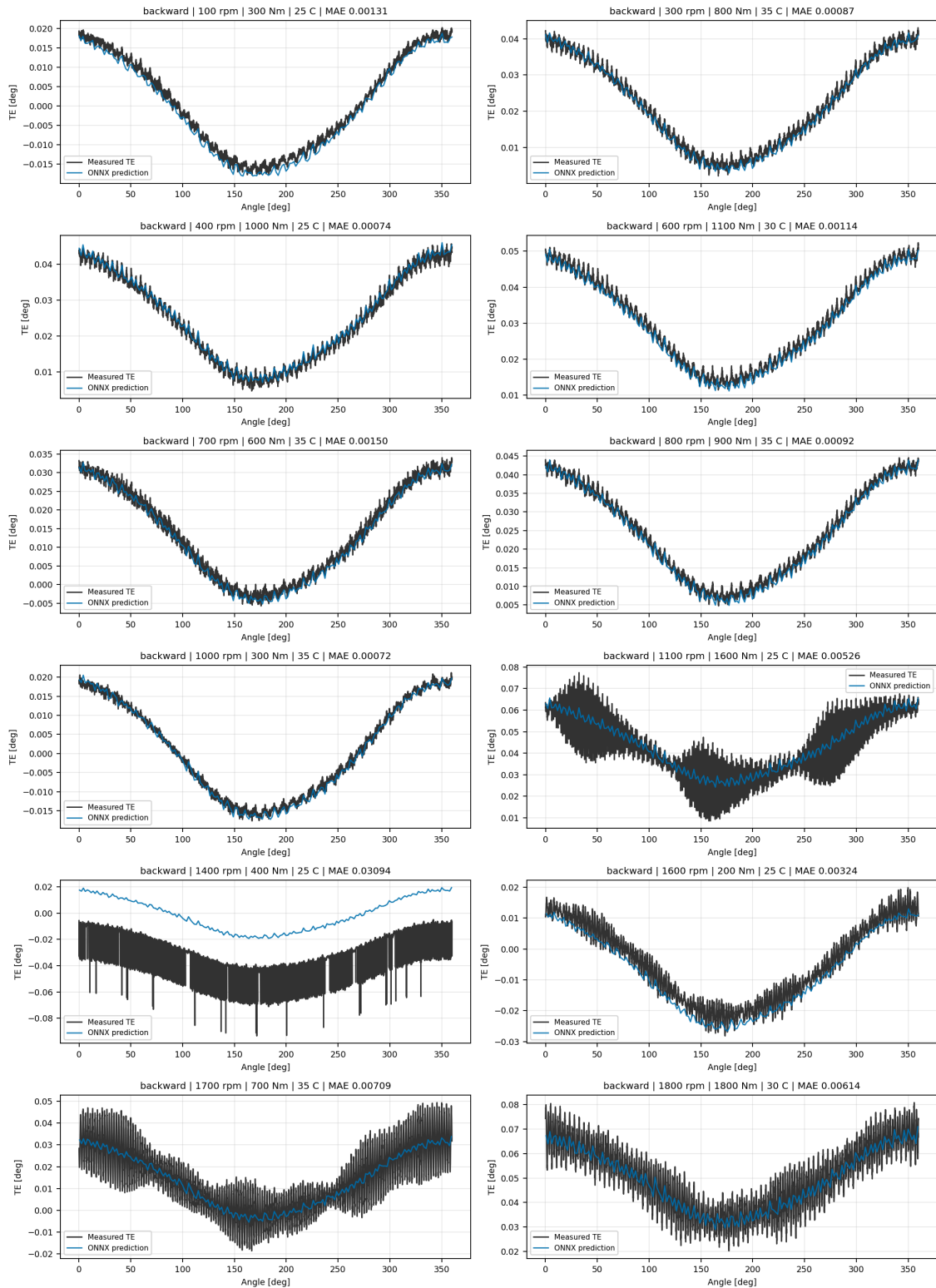
Polished Dataset + Setpoints | wave3_3_raw_centered_shape_curve_aware | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

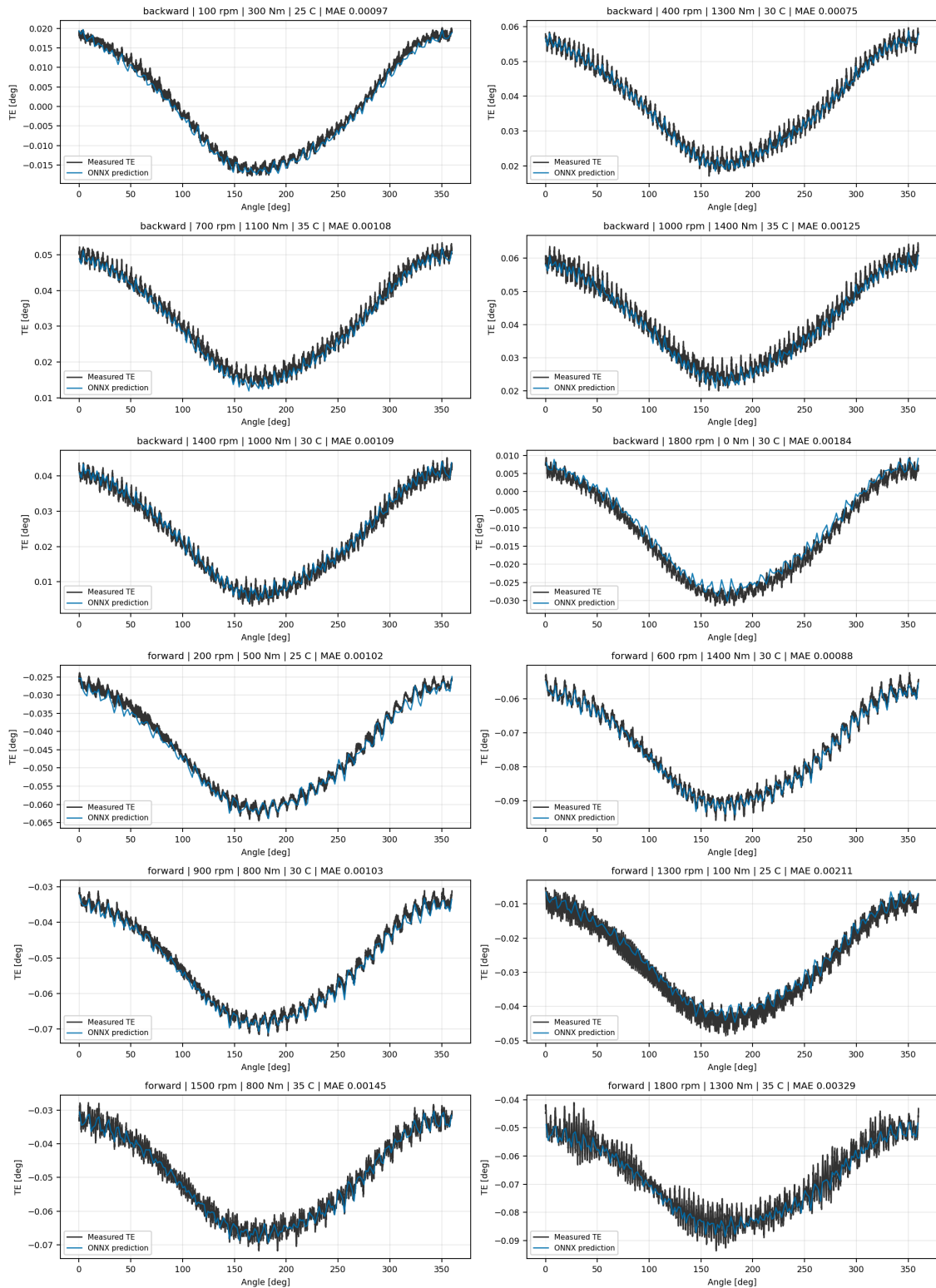
Polished Dataset + Setpoints | wave3_3_raw_centered_shape_curve_aware | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | wave3_3_raw_centered_shape_curve_aware | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: wave3_3_raw_centered_shape_curve_aware ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave3_3_raw_centered_shape_curve_aware_fw_polished_actual_values	2026-07-11-20-11-00__te_wave3_3_raw_centered_shape_curve_aware_fw_polished_actual_values	polished_point_v1
backward	te_wave3_3_raw_centered_shape_curve_aware_bw_polished_actual_values	2026-07-11-20-41-54__te_wave3_3_raw_centered_shape_curve_aware_bw_polished_actual_values	polished_point_v1
global	te_wave3_3_raw_centered_shape_curve_aware_global_polished_actual_values	2026-07-11-19-14-45__te_wave3_3_raw_centered_shape_curve_aware_global_polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/wave3_3_raw_centered_shape_curve_aware/forward/2026-07-11-20-11-00__te_wave3_3_raw_centered_shape_curve_aware_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave3_3_raw_centered_shape_curve_aware/forward/2026-07-11-20-11-00__te_wave3_3_raw_centered_shape_curve_aware_fw_polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=095-val_mae=0.00193478.ckpt
backward	models/polished_dataset/actual_values/exported/wave3_3_raw_centered_shape_curve_aware/backward/2026-07-11-20-41-54__te_wave3_3_raw_centered_shape_curve_aware_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave3_3_raw_centered_shape_curve_aware/backward/2026-07-11-20-41-54__te_wave3_3_raw_centered_shape_curve_aware_bw_polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=186-val_mae=0.00185514.ckpt
global	models/polished_dataset/actual_values/exported/wave3_3_raw_centered_shape_curve_aware/global/2026-07-11-19-14-45__te_wave3_3_raw_centered_shape_curve_aware_global_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave3_3_raw_centered_shape_curve_aware/global/2026-07-11-19-14-45__te_wave3_3_raw_centered_shape_curve_aware_global_polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=210-val_mae=0.00182815.ckpt

Aggregate Metrics

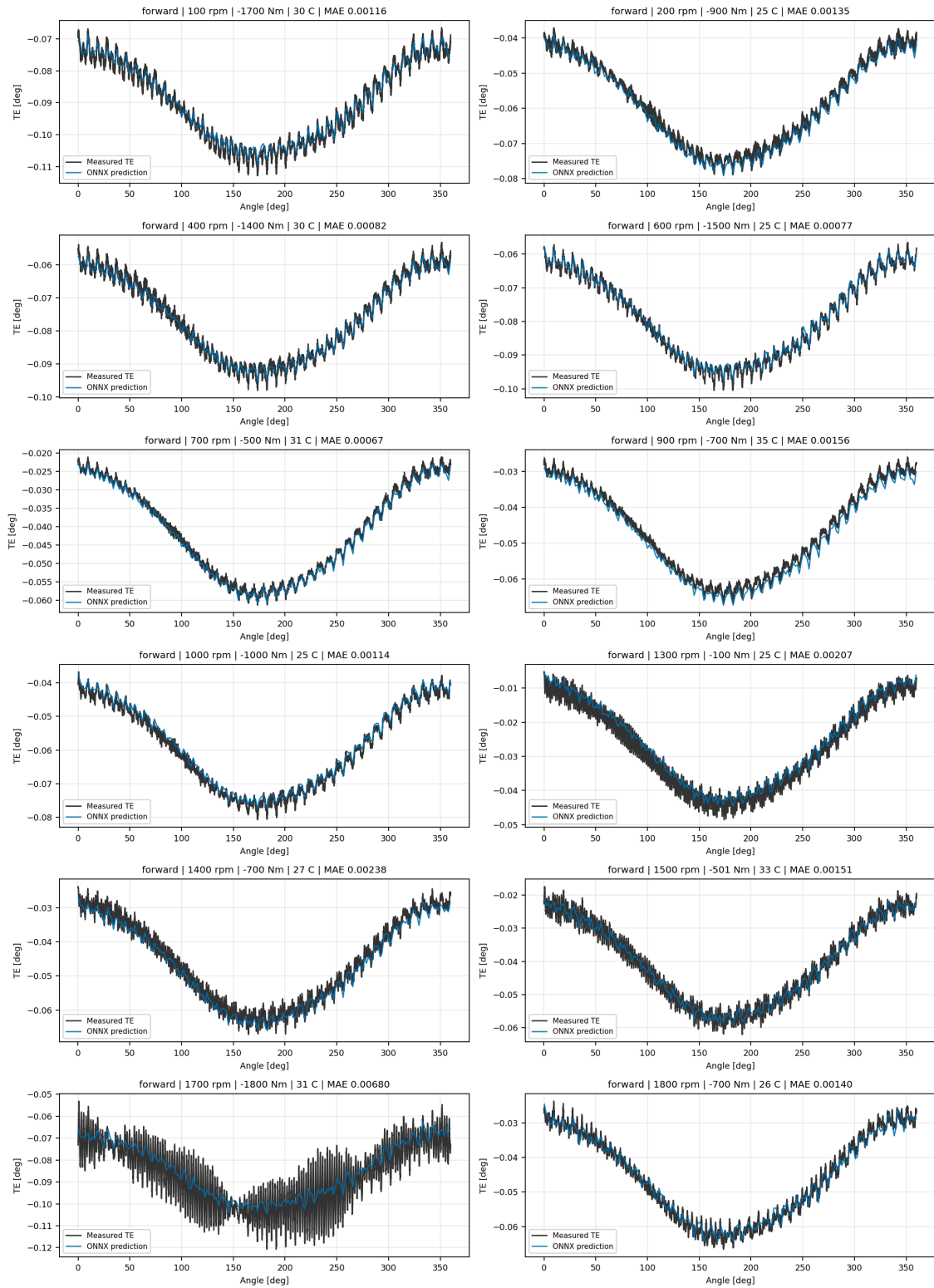
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001926	0.002295	4.247	10.410
backward	94	0.002197	0.002650	4.176	10.447
global	194	0.001967	0.002371	3.979	10.091

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000458	0.000946	0.001571	0.004146
backward	-0.000006	0.000575	0.002081	0.005380
global	-0.000033	0.000664	0.001801	0.004572

Forward 12-Curve Page

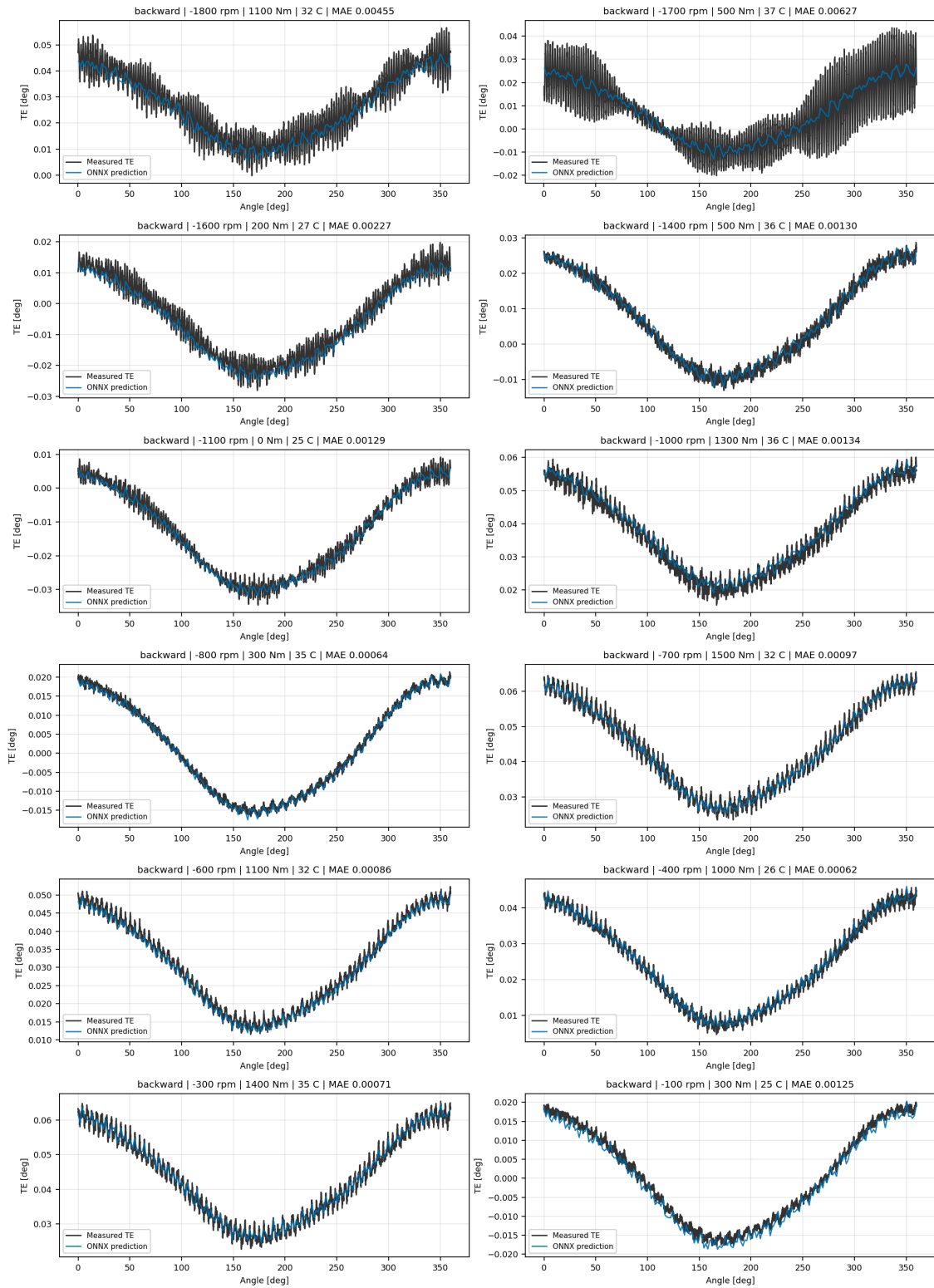
Polished Dataset + Actual Values | wave3_3_raw_centered_shape_curve_aware | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

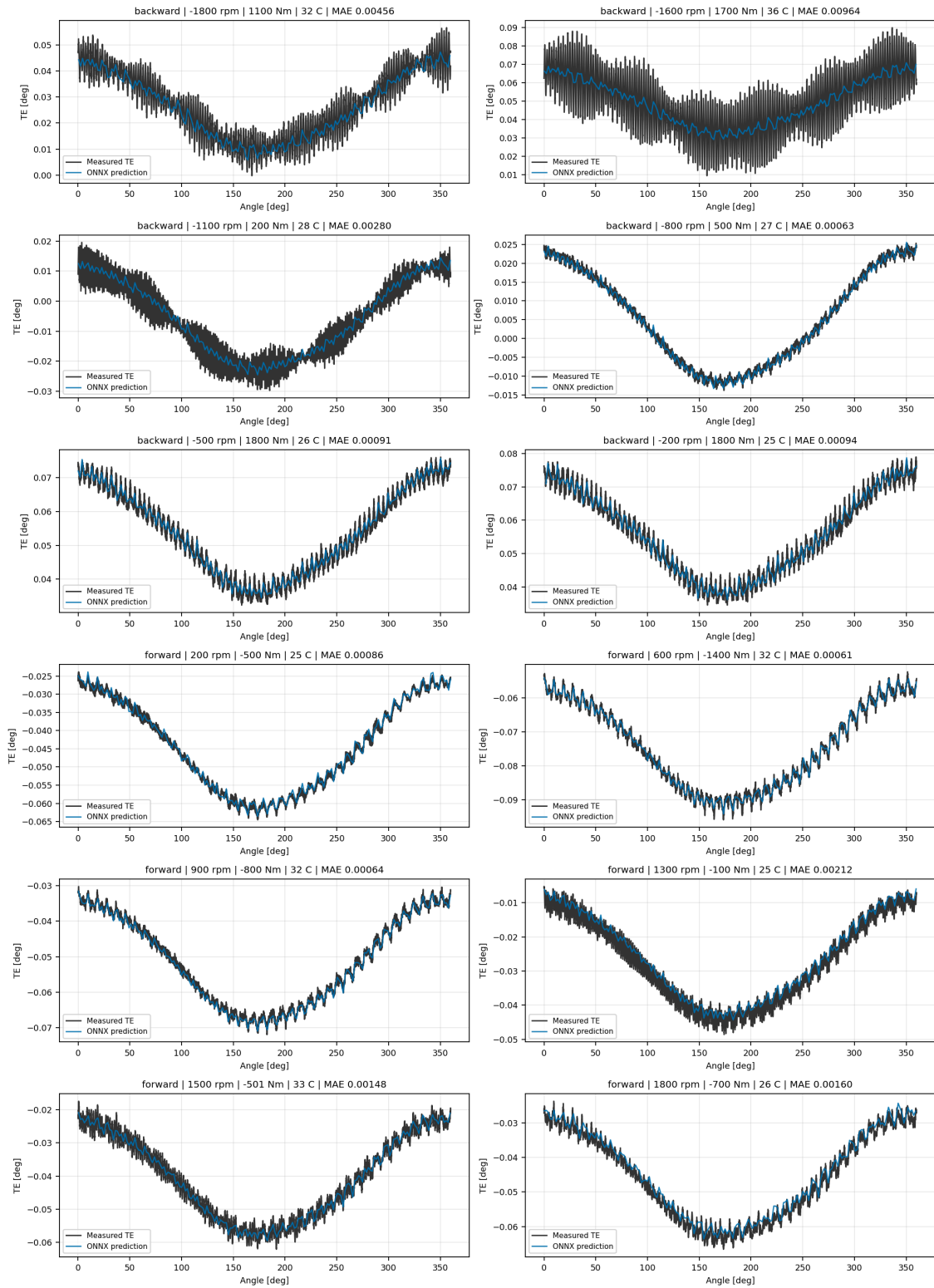
Polished Dataset + Actual Values | wave3_3_raw_centered_shape_curve_aware | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | wave3_3_raw_centered_shape_curve_aware | global



polished_dataset_actual_values global 12-curve collage